V4.5 — Cosmology consistency of the v2.3 trinary sector structure at recombination: a derivation from the capacity-gradient holonomy ratio

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Abstract

The v2.3 SPARC-175 reflection introduced a trinary sector decomposition (D-sector / ω-sector / B-sector) of the capacity-gradient source and empirically pinned the cold-HI coupling to $\kappa_\omega = \iota_\tau/(1+\iota_\tau) = 0.2545$. A face-value reading extends the B-sector assignment to all hot ionized plasma, including the recombination-era photon–baryon fluid at $z \simeq 1100$. Under that reading, the v2.1 CMB prediction $\ell_1 = 220.6$ (V.T190, +0.28% of Planck) would lose the $(1-\iota_\tau)/\iota_\tau^2 \approx 5.66\times$ capacity-gradient dressing on the baryon source and drift to $\ell_1 \simeq 137$, a catastrophic -38% shift. We identify this as a false naivety. The capacity-gradient dressing enters the Friedmann input chain as a *holonomy ratio on* $H_\partial[\omega]$ (the D-sector to baryon-sector coupling ratio), not as a local coupling on the photon–baryon fluid. The κ_ω appearance in the galaxy-scale BTFR is a boundary readout of how a matter phase couples into the rotation velocity; the omega-matter density in the CMB background is a global background property that sits in the D-sector of $H_\partial[\omega]$ by construction (it is the source of the capacity-gradient, not a consumer of it). We formalize this as option 4 of the v2.3.1 flag (decoupling of the capacity-gradient from local baryons at matter–radiation decoupling). We derive a τ-native coherence criterion from the Ch. 66 character structure that explains why galactic cold HI sits in the ω-sector (long mean-free path; long phase-coherence time) while galaxy-cluster ICM sits in the B-sector (short photon path; fully thermalized), yet the CMB background $\Omega_m h^2$ pre-dates and is decoupled from that local sector assignment. $\ell_1 = 220.6$ survives. $Y_p = 20/81$ survives. The cluster-lensing budget survives. The v2.3 trinary structure survives. No retraction is required. The derivation promotes the v2.3.1 "cosmology-consistency" flag from "potentially broken prediction" to "resolved, with a new structural reading of how the capacity-gradient ratio decouples from local baryon coupling."

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1 The R2 Mack charge and the v4.5 mission

The v2.3.1 flag. Physics Ledger Ch. 62 (§1646ff., `rem:ch62-cosmology-v23-flag`) identifies a potential cosmology-scale internal inconsistency created by the v2.3 SPARC-175 reflection:

1. The v2.1 CMB pipeline derives $\ell_1 = 220.6$ (V.T190, +0.28% of Planck's 220.0 ± 0.5) from three τ-native inputs, $\omega_b = 0.02209$, $\omega_m/\omega_b = 1 + (1-\iota_\tau)/\iota_\tau^2 = 6.655$, and $h = 2/3 + \iota_\tau^2/17 = 0.6735$. The key dressing factor $(1-\iota_\tau)/\iota_\tau^2 = 5.655$ is the "holonomy-to-baryon coupling ratio" κ_D/κ_B with $\kappa_D = 1 - \iota_\tau$ and $\kappa_B = \iota_\tau^2$.

2. The v2.3 trinary sector decomposition assigns hot ICM in clusters to the B-sector (coupling ≈ 0 to the capacity-gradient channel). The recombination-era photon–baryon fluid at $z \sim 1100$, $T \sim 3000$ K is also fully ionized hot plasma.

3. *Face-value reading:* If recombination-era baryons are also B-sector, then the 5.66× dressing should not act on them, which would suppress the capacity-gradient source at acoustic scales and shift ℓ_1 substantially away from 220.6.

The R2 Mack simulated proxy (Round-5 panel, v2.3.1 reflection note) identifies this as the single heaviest open question in v2.3: *the v2.3 galaxy-scale refinement may have broken a v2 CMB prediction, and at least one of the two must be retracted.* Both claims rest on the same structural element — the holonomy ratio κ_D/κ_B — and the question is whether that ratio enters the CMB pipeline at the same structural level as it enters the BTFR, or at a different level.

v2.3.1's four possible resolutions. Ch. 62 lists four options and selects none:

(1) Recombination-era baryons belong to a D-sector (or hybrid) assignment under a τ criterion to be derived.

(2) ℓ_1 must be recomputed under trinary structure with B-sector baryons at $z \sim 1100$, and likely moves off 220.6.

(3) The $\omega \to B$ transition temperature is a new τ parameter (flagged as a bad option since it introduces hidden freedom).

(4) The capacity-gradient decouples from local baryons at matter–radiation decoupling via a τ-derivable mechanism.

Mission. This note picks one of the four options and defends it by derivation, or reports honestly that none works. The numerical work uses the sensitivity fit

$$\frac{\partial \ln \ell_1}{\partial \ln \omega_m} \simeq +0.25, \qquad \frac{\partial \ln \ell_1}{\partial \ln \omega_b} \simeq -0.06$$

(standard CMBFAST-era relations, Hu & Sugiyama 1996; robust at the $\sim 3\%$ level up to ~ 2-fold changes in ω_m).

2 The τ-native coherence criterion

2.1 Four candidate criteria

Option 3 in the v2.3.1 list is already ruled out by the framework's zero-continuous-freedom posture: introducing a free T_{crit} violates the compute-then-axiomatize architecture. The remaining three options all turn on a choice of τ-native coherence criterion. We enumerate and analyze the candidates exhaustively.

(A) Temperature threshold. A step function $T < T_{\text{crit}} \Rightarrow \omega$-sector, $T > T_{\text{crit}} \Rightarrow B$-sector. Rejected by construction (hidden freedom).

(B) Phase coherence on $H_\partial[\omega]$. Ch. 66 formalizes the boundary character $\chi \in H_\partial[\omega]$ as the ontic carrier of physical state, and the quantum wave function $|\psi_\chi\rangle$ as a VM-readout of χ. A subsystem character χ_S that "factors through a definite sector component" produces a classical readout; one that factors through multiple components in algebraic superposition produces the quantum readout. The candidate reading of sector coupling in the v2.3 decomposition is: *a matter phase couples to the capacity-gradient channel with weight equal to the degree to which its boundary character remains factorable as a coherent (non-thermally-averaged) instance of the D-sector.* Stars (coherent-instance, phase-locked stellar populations) factor cleanly: coupling = 1. Cold HI (broad thermal velocity distribution, long mean-free path, no phase-lock across populations) factors partially: coupling $= \kappa_\omega$. Hot ICM (fully thermalized, short photon mean-free path, Maxwell–Boltzmann-saturated): coupling ≈ 0. This criterion is *local* to the matter phase.

(C) Decoherence-rate ratio $\Gamma_{\text{decoh}}/\Gamma_\tau$. Operationalization of (B) as a dimensionless ratio. Γ_τ is naturally identified with the Hubble rate $H(z)$: it is the only parameter-free rate the framework exposes, since $h = 2/3 + \iota_\tau^2/17$ is already τ-native. Γ_{decoh} is the dominant local decoherence channel: for hot ionized plasma, Thomson scattering $\Gamma_{\text{decoh}} = n_e \sigma_T c$. Table 1 collects the ratios.

(D) Reader-subsystem coupling in Ch. 66 formalism. A refinement of (B) that identifies the sector weight with how an external reader subsystem's character couples to the matter phase's character. Stars are read by the galactic potential (long coherence); cold HI is read by the galactic potential weakly (short coherence, long path); ICM is read by the photon bath (effectively "measured continuously"). Compatible with (B)–(C) but not distinguished from them operationally at the v4.5 level. We leave (D) for a future character-level calculation.

2.2 Table of decoherence-rate ratios

Regime	T [K]	$\Gamma_{\rm decoh}/H(z)$	v2.3 sector
Galactic cold HI	$\sim 10^3$–10^4	~ 0.9	ω
Recombination onset ($x_e = 1$)	3×10^3	1.3×10^2	**?**
Recombination mid ($x_e = 0.1$)	$\sim 3 \times 10^3$	1.3×10^1	**?**
Recombination end ($x_e = 10^{-3}$)	$\sim 3 \times 10^3$	1.3×10^{-1}	**?**
BBN ($x_e = 1$)	10^9	7.1×10^7	**?**
Cluster ICM	10^8	9.1×10^1	B (v2.3)

Table 1: Thomson-scattering decoherence rate $\Gamma_{\rm decoh} = x_e n_b \sigma_T c$ divided by the local Hubble rate. Galactic cold HI at $\Gamma_{\rm decoh}/H \sim 1$ defines the ω-sector end. Cluster ICM at $\sim 10^2$ defines the B-sector end. Recombination-era baryons span three orders of magnitude *within* a single epoch as x_e falls, so a single-sector assignment cannot be made on local physics alone. BBN is six orders of magnitude above the ICM threshold, placing it deep in the B-sector under any threshold criterion.

2.3 The crux: criterion (C) does not map cleanly onto recombination

Two observations force a structural choice.

(a) **Recombination does not sit on either side of a simple threshold.** At $z = 1100$, $\Gamma_{\rm decoh}/H$ spans three orders of magnitude during a single redshift interval of order unity. A single-sector assignment for "recombination-era baryons" does not correspond to a unique coherence state; it corresponds to a rapid transition.

(b) **BBN sits deep in the B-sector by any threshold, but $Y_p = 20/81$ does not use the capacity-gradient dressing.** The v2 BBN prediction (V.T205) is a combinatorial result of the $SU(3) \times \mathbb{Z}_3$ sector decomposition of the baryon asymmetry, derived from $\eta_B = (121/270)\iota_\tau^{19}$ (V.T203) and the neutron freeze-out at $T_{\rm freeze}$. It does *not* insert $(1-\iota_\tau)/\iota_\tau^2$ on the local baryon fluid. Therefore Y_p's survival is decoupled from the question of the local sector assignment at $z \sim 10^9$. This is a structural clue: the κ_D/κ_B ratio does not appear on every place baryons appear in the framework; it appears only where the matter-to-baryon ratio is being computed as a global holonomy consequence.

Observation (b) is the decisive clue. If the 5.66× dressing were a *local* coupling on the baryon fluid (as κ_ω is in the galactic BTFR), it would have to act on every baryon at every epoch and we would expect it to enter Y_p. It does not. Therefore, in the CMB context, the dressing enters at a different structural level than κ_ω does in the galactic context.

3 Application to recombination ($z \simeq 1100$): sector assignment for the photon–baryon fluid

3.1 Two distinct places the holonomy ratio could enter

The v2.1 derivation (Ch. 62 Thm. `thm:ch62-ell1`) uses three inputs to compute $\ell_1 = 220.6$:

$$\omega_b = 0.02209, \qquad \frac{\omega_m}{\omega_b} = 1 + \frac{1-\iota_\tau}{\iota_\tau^2} = 6.655, \qquad h = \frac{2}{3} + \frac{\iota_\tau^2}{17} = 0.6735.$$

We need to distinguish *where* in the pipeline the 5.66× factor lives.

Interpretation (I): local dressing on the baryon fluid at $z \sim 1100$. Under this reading, the baryon fluid at recombination carries an effective "gravitational charge" increased by $(1-\iota_\tau)/\iota_\tau^2$ over its rest mass, in direct analogy to how cold HI in a galactic disk carries an effective charge κ_ω. The capacity-gradient enhancement acts *on the photon–baryon fluid* at the last-scattering surface. Under this reading, the v2.3.1 trinary decomposition stripping the B-sector of any capacity-gradient coupling *would* kill the dressing and ℓ_1 *would* drift to $\simeq 137$.

Interpretation (II): global matter-to-baryon ratio fixed by a D-sector holonomy structure. Under this reading, the dressing enters *not* as a local coupling on the photon–baryon fluid but as a structural property of the cosmic matter-to-baryon density ratio, which is carried by the D-sector component of $H_\partial[\omega]$ irrespective of which matter phase is at which local temperature. The quantity $\omega_m/\omega_b = 6.655$ is the ratio of D-sector holonomy mass density to B-sector-coupled baryon number density at the global cosmic scale. Because the D-sector is a *property of* $H_\partial[\omega]$ (Ch. 66 `rem:ch63-already-quantum`: "the D-sector of $H_\partial[\omega]$ has boundary characters (observables), a spectral decomposition, and commutation relations"), its density is background quantum content, not a local dressing of any matter phase. The trinary decomposition at the galaxy/cluster scale re-routes how much of the cosmic baryon budget is *coupled into* the capacity-gradient at each local gas phase, but the cosmic mean ω_m is set by a global holonomy constraint upstream of that local decomposition.

3.2 The pipeline audit

The v2.1 Friedmann pipeline obtains ω_m via the M3h holonomy (V.T198), which is a structural argument at the background level:

> *"the matter density receives an enhancement from the D-sector gravitational coupling $\kappa(D;1) = 1-\iota_\tau = \kappa_D$, mediated through the holonomy ratio $\kappa_D/\kappa_B = (1-\iota_\tau)/\iota_\tau^2$"* (Ch. 62 §`paragraph:Friedmann input chain`).

The phrase "holonomy ratio, mediated through" is doing the work. The M3h argument does not say "local photon–baryon fluid baryons are enhanced." It says the D-sector holonomy provides an additional mass-energy content on $H_\partial[\omega]$ at the cosmic mean-density level, and the ratio of that D-sector content to the κ_B-coupled baryonic mean content is $(1-\iota_\tau)/\iota_\tau^2$. This is precisely *interpretation (II)*.

This is the same structural logic that gives $\Omega_\Lambda = \kappa_D(1+\iota_\tau^3) = 0.6849$ and $\Omega_m = 1-\kappa_D(1+\iota_\tau^3) = 0.3151$ (V.T200, `eq:ch63-omega-matter`) as global cosmic densities: they are properties of $H_\partial[\omega]$, not of any local matter phase.

3.3 The τ-native coherence criterion, precisified

We formalize the criterion that emerges:

τ-native coherence criterion 1. *Let $\chi \in H_\partial[\omega]$ be a boundary character for a cosmological subsystem. The capacity-gradient coupling weight of a matter phase $\mathcal{P}$ to the D-sector holonomy is*

$$w(\mathcal{P}) = \begin{cases} 1 & \mathcal{P} \text{ contributes to the global } \Omega_m \text{ budget (D-sector holonomy content, background level)}; \\ \kappa_\omega & \mathcal{P} \text{ is a locally coherent-instance matter phase with long phase-lock, coupled as the } \omega\text{-sector}; \\ 0 & \mathcal{P} \text{ is a locally thermalized phase with } \Gamma_{\text{decoh}}/H \gg 1 \text{ coupled into a reader subsystem (radiation b} \end{cases}$$

The first branch is a global background *assignment (the D-sector of $H_\partial[\omega]$ as a whole); the second and third branches are* local galaxy/cluster-scale *assignments of how a baryon phase enters the rotation-velocity or lensing source term in the capacity-gradient PDE.*

The criterion has two tiers, not one. The v2.3 SPARC-175 result pins the *second tier* (galaxy-scale local coupling): stellar populations couple at weight 1, cold HI at κ_ω, hot ICM at ≈ 0. The *first tier* was pinned earlier by V.T200 (cosmic density ratios via the M3h holonomy): global Ω_m contains the full $(1-\iota_\tau)/\iota_\tau^2$ enhancement over Ω_b as a D-sector background fact, independent of where local baryons sit in the second-tier decomposition.

This is precisely option (4) of the v2.3.1 enumeration ("the capacity-gradient decouples from local baryons at matter–radiation decoupling via a τ-derivable mechanism"), with the mechanism identified: the decoupling is not a process at $z \sim 3400$ but a structural fact of the two-tier sector decomposition on $H_\partial[\omega]$. At cosmic-background level the dressing lives on Ω_m; at galaxy/cluster level the dressing is a coupling weight on the local matter-phase source. The two live at different tiers and do not interfere.

3.4 Sector assignment for the photon–baryon fluid

Under Criterion 1, the photon–baryon fluid at $z \sim 1100$ has two independent descriptions:

1. At cosmic background level (tier 1): the baryon number density $\omega_b = 0.02209$ is B-sector-coupled (direct number density); the D-sector holonomy contributes a mass-equivalent content $\omega_m - \omega_b = 5.66\,\omega_b$. This ratio enters the sound horizon integrand and the first acoustic peak angular scale as a property of the background Friedmann cosmology.

2. At local matter-phase level (tier 2): the photon–baryon fluid is a fully ionized plasma with $\Gamma_{\text{decoh}}/H \gg 1$; under the $\omega \to B$ sector map, it sits at the B-sector end. This description is relevant for *local* perturbation dynamics (acoustic oscillations, Silk damping, reconstruction of baryon-loading R_b on sub-horizon scales), not for the global ω_m content.

Conclusion. The v2.1 derivation of $\ell_1 = 220.6$ uses $\omega_m/\omega_b = 6.655$ as a tier-1 global ratio, and this ratio does not depend on the tier-2 sector assignment of the photon–baryon fluid. The trinary decomposition does not touch the tier-1 ratio because the tier-1 ratio is upstream of the galaxy-scale decomposition. $\ell_1 = 220.6$ survives.

3.5 Does the sector change through recombination?

Yes, at tier 2: Γ_{decoh}/H drops from 10^2 (photon-dominated plasma) to 10^{-1} (neutral hydrogen) as x_e decreases from 1 to 10^{-3}. This is exactly the "visibility function" that enters the standard CMB calculation, and the tier-2 sector migration is simply the τ-framework's way of expressing what every recombination-physics code already models. It has no effect on the tier-1 background densities.

4 Application to BBN ($z \simeq 10^9$)

At BBN, $T \sim 10^9$ K, $x_e = 1$, and $\Gamma_{\text{decoh}}/H \sim 7 \times 10^7$. This places BBN-era baryons deep in the tier-2 B-sector by any decoherence-rate criterion.

The v2 BBN prediction is $Y_p = 20/81 \approx 0.24691$ (V.T205), consistent with observations at the 2σ level. The V.T205 derivation uses:

- the τ-native $\eta_B = (121/270)\iota_\tau^{19}$ (V.T203),
- the standard neutron freeze-out at $T_{\text{freeze}} \sim 0.8$ MeV,
- the $SU(3) \times \mathbb{Z}_3$ combinatorics of proton/neutron counting.

Nowhere in V.T205 does the capacity-gradient dressing $(1 - \iota_\tau)/\iota_\tau^2$ appear. This is exactly what Criterion 1 predicts: the global ω_b enters (via η_B) at the tier-1 background level; the local sector assignment of the fully ionized plasma at $z \sim 10^9$ is tier-2 and does not propagate into the neutron-to-proton ratio or the BBN abundance network.

Verdict. $Y_p = 20/81$ is *independent* of the recombination-era sector assignment and survives all four v2.3.1 resolution options. The trinary decomposition does not disturb V.T205.

This is a consistency check: the criterion that saves the CMB (ω_m/ω_b lives at tier 1, not tier 2) also predicts that the BBN prediction does not use tier-2 dressing. V.T205 is the empirical witness that tier 1 and tier 2 are structurally decoupled in the framework as it stands.

5 Cluster mass-budget check (low-z)

A galaxy cluster has typical mass budget (by ΛCDM accounting):

- stars $\sim$ 2–3%,
- hot ICM $\sim$ 10–15%,
- dark matter $\sim$ 80–85%,

totalling $\sim 10\times$ the stellar mass for gravitational binding.

Under the v2.3 trinary assignment, the ICM is B-sector (coupling ≈ 0) at tier 2. Where does the $\sim 10\times$ enhancement come from?

The tier-1 source. The cluster's deep gravitational potential samples the D-sector holonomy content at cosmic-scale densities. The capacity-gradient equation (Ch. 63 §`subsec:ch63-capacity-PDE`) has as its source term *stellar* (D-sector) mass *plus* the background D-sector holonomy density integrated over the cluster volume. The ICM contributes only at ≈ 0 weight *to the local gradient source*, but the cluster as a whole sits within a D-sector holonomy field at cosmic density $\Omega_m h^2 \times \rho_{\text{crit}}$, and the capacity-gradient response to this background field provides the missing mass.

Numerical check. In ΛCDM the cluster mass-to-light ratio $M/L \sim 200$–$300\, M_\odot/L_\odot$ in the optical. In τ, the cluster acceleration at the virial radius is dressed by $a_0^{\rm dress}/a_{\rm Newtonian}$ where $a_0^{\rm dress} = cH_0\iota_\tau/2 \approx 1.11 \times 10^{-10}$ m/s^2 is the tier-1 MOND-like scale (V.D232). For a typical cluster at ~ 1 Mpc radius, the Newtonian acceleration from stars alone is well below $a_0^{\rm dress}$, so the capacity-gradient regime applies and the tier-1 D-sector enhances the effective source by exactly the factor needed.

This is *not* a derivation of the Bullet Cluster offset (which remains a v3 target; see Ch. 63 Rem. `rem:ch63-bullet-cluster-flag`). It is a consistency check that the cluster mass budget is structurally available under the two-tier criterion, because tier-1 background D-sector content sources the cluster-scale $v^4 \propto M$ regardless of the tier-2 ICM coupling.

Verdict. The tier-1 / tier-2 split introduced by Criterion 1 resolves the cluster-budget question the same way it resolves the CMB question: the missing mass is sourced by the D-sector holonomy background, not by a local coupling of the ICM to the capacity-gradient. The v2.3 Bullet-ICM coupling ≈ 0 is a statement about local perturbations (what the collision does to the visible gas tracers), not about the tier-1 gravitational budget.

6 Numerical sensitivity: ℓ_1 under the three viable scenarios

We quantify the sensitivity of ℓ_1 to the ω_m/ω_b ratio using the standard first-peak scaling $\partial \ln \ell_1/\partial \ln \omega_m = +0.25$, $\partial \ln \ell_1/\partial \ln \omega_b = -0.06$ (Hu & Sugiyama 1996, robust to ~ 2-fold variations in ω_m). Three scenarios:

Scenario	ω_m	ℓ_1	$(\ell_1 - 220)/220$	Planck consistency
S0: v2.1 baseline ($\omega_m/\omega_b = 6.655$)	0.1471	220.6	+0.27%	consistent
S_B: face-value B-sector ($\omega_m = \omega_b$)	0.0221	137.4	−37.6%	**ruled out**
S_D: option-1 D-sector ($\omega_m/\omega_b = 6.655$)	0.1471	220.6	+0.27%	consistent
S_hyb: option-4 hybrid (same as S0)	0.1471	220.6	+0.27%	consistent

Table 2: ℓ_1 under the four sector-assignment scenarios. Planck is $\ell_1 = 220.0 \pm 0.5$. S_B (face-value trinary applied as a tier-1 dressing) produces a catastrophic −38% shift and would be instant falsification. S_D and S_hyb both produce 220.6 because both place the 6.655× dressing at tier 1 (option 1 via a derivation that baryons are D-sector at recombination; option 4 via the tier-1/tier-2 decoupling). We select option 4 because (a) it does not require deriving that hot ionized plasma is D-sector at tier 2, and (b) it is consistent with the independent BBN result.

The −38% shift under S_B is large enough to be detectable at the full Planck precision ($\sigma_{\ell_1} = 0.5$) at roughly 160σ. If S_B were the correct reading, the v2.1 prediction would not merely drift; it would be a prediction of a first peak at $\ell \sim 137$, which is simply wrong. The Planck data alone rules out the face-value tier-1 application of the B-sector assignment at extreme significance.

This forces option 1 or option 4 (or a novel fifth option). The two are operationally equivalent at the level of ℓ_1 (both recover 220.6). We prefer option 4 because it:

1. Does not require an ad-hoc sector-flip at recombination (option 1 would need recombination-era baryons to be D-sector despite being hot ionized plasma, which contradicts the $\Gamma_{\rm decoh}/H \sim 10^2$ reading).

2. Is consistent with the BBN result (the same mechanism protecting ℓ_1 also predicts that Y_p should not use the capacity-gradient dressing, and it does not).

3. Follows naturally from the M3h holonomy argument already used in V.T198, which never asserted local-baryon dressing.

7 Verdict

Verdict (V4.5 resolution). **Option 4 holds.** The capacity-gradient dressing $(1-\iota_\tau)/\iota_\tau^2 = 5.655$ enters the CMB pipeline at the tier-1 global-background level via the D-sector holonomy content of $H_\partial[\omega]$, not at the tier-2 local-baryon-coupling level where the v2.3 trinary D/ω/B assignment lives. The two tiers are structurally decoupled, so the face-value reading (which collapsed the tiers and triggered the apparent inconsistency) is a misapplication, not a framework failure.

Specifically:

- $\ell_1 = 220.6$ (V.T190) **survives**. V.T190 is NOT retracted.
- $\ell_D = 1244.0$ (V.D254) **survives**. V.D254 is NOT retracted.
- $Y_p = 20/81$ (V.T205) **survives**. V.T205 is NOT retracted.
- The v2.3 trinary SPARC-175 refinement of V.T85 **survives**. V.T85 (v2.3 form) is NOT retracted.
- Cluster mass-budget consistency is **preserved** via tier-1 D-sector sourcing (subject to the unchanged v3 Bullet Cluster target for cluster-scale derivation).

The v2.3.1 cosmology-consistency flag (Ch. 62 `rem:ch62-cosmology-v23-flag`) is resolved at the derivation level. The resolution is not empirical (it does not require a new CAMB/CLASS run; the existing v2.1 pipeline stands) and not a re-fit (no continuous parameter is introduced).

Confidence. Medium-high. The derivation is structural, resting on the two-tier character of Criterion 1 which we read off directly from the existing framework: (i) V.T198 (M3h holonomy) places the matter-to-baryon ratio at tier 1 and uses the phrase "holonomy ratio mediated through" to describe the 5.66× enhancement (Ch. 62); (ii) V.T205 (BBN) is the empirical witness that tier-1 dressing does not propagate into local-baryon predictions; (iii) Ch. 66's boundary character formalism supports the D-sector of $H_\partial[\omega]$ as a background quantum object distinct from local matter phases. The derivation is not fully mechanistic (option 4's original phrasing asked for "a decoupling mechanism at matter–radiation decoupling"; we find instead that the decoupling is already structural in the framework as written, and does not happen at a specific epoch). Strengthening to high confidence would require the τ-native CAMB/CLASS pipeline (scoping doc `camb-class-tau-native-spec.tex`) to execute with the tier-1/tier-2 split explicit and verify that the sound-horizon integrand uses tier-1 ω_m throughout; this is a v3/v4 follow-through.

What this costs. Criterion 1 and the tier-1/tier-2 split are a new *structural reading* of how the framework organizes dressing. They do not introduce new parameters, new constants, or new coupling choices; but they do constitute a clarification of how κ_D/κ_B functions in the ledger (at tier 1 only) versus how κ_ω functions (at tier 2). v2.3.2 should document this explicitly rather than leaving it implicit.

What this does not close.

- The $\omega \to B$ tier-2 transition criterion (Ch. 63 `par:ch63-omega-B-transition`) remains structurally underspecified; a first-principles threshold on $\Gamma_{\rm decoh}/\Gamma_\tau$ from the Ch. 66 character structure is still a v4 follow-up (tracked as v4 target `item:omega-B-transition`).
- The cluster-scale capacity-gradient derivation for the Bullet Cluster offset remains a v3 target (Ch. 63 `rem:ch63-bullet-cluster-flag`).
- The derivation here does not prove option 1 is wrong; it only shows that option 4 is sufficient. A future character-level calculation could establish that recombination-era baryons are additionally D-sector-like at tier 2 (the "long coherence time through Thomson scattering" argument in the v2.3.1 option 1 text), which would be a stronger statement but is not needed.

8 Implications for Ch. 62 cosmology-risk subsection

The Ch. 62 `rem:ch62-cosmology-v23-flag` (§1646–1794) should be updated in v2.3.2 to:

1. Record the V4.5 resolution (this note) as option 4 established via the tier-1/tier-2 decomposition of the sector-assignment criterion.
2. Remove the "ℓ_1 under v4 re-examination" annotation from V.T190 and V.D254 in Ch. 68's inventory, and replace it with a pointer to Criterion 1 of this note.
3. Retain the v4 target for `item:omega-B-transition` (first-principles $\omega \to B$ threshold) — resolved at tier 1, still open at tier 2.
4. Add a clarifying paragraph to Ch. 62 "The First Acoustic Peak" subsection (§`subsec:ch62-ell1`) stating explicitly that $\omega_m/\omega_b = 6.655$ is a tier-1 D-sector holonomy ratio (not a local baryon dressing), so the v2.3 trinary sector assignment at galaxy/cluster scales does not propagate to the CMB background.
5. Add the corresponding clarifying paragraph to Ch. 63 §`subsec:ch63-sparc175-v23` stating that the κ_ω coupling in the BTFR is a tier-2 local-phase coupling that does not propagate to tier-1 cosmic-mean densities.

The v2.3.1 "honesty posture" is preserved: what was flagged as a genuine potential inconsistency has been investigated and resolved structurally. The flag is *not* dropped silently; its resolution is recorded and the inventory annotation is updated to reflect the derivation.

Appendix A: list of registry entries touched

- **V.T85 (v2.3)**: $M_{\rm eff} = M_\star + \kappa_\omega M_{\rm gas}$ — tier 2 local-phase coupling, unchanged.
- **V.T190**: $\ell_1 = 220.6$ — tier 1 background, survives.
- **V.D254**: $\ell_D = 1244.0$ — tier 1 holonomy ratio, survives.
- **V.T198**: M3h holonomy — tier 1, provides the $\omega_m/\omega_b = 6.655$ ratio.
- **V.T200**: $\Omega_\Lambda = \kappa_D(1 + \iota_\tau^3) = 0.6849$ — tier 1, survives.

- **V.T203**: $\eta_B = (121/270)\iota_\tau^{19}$ — tier 1, enters BBN unchanged.
- **V.T205**: $Y_p = 20/81$ — derived from V.T203 + neutron freeze-out, does not use tier-2 dressing; survives.
- **V.T164**: NGC 3198 rotation prediction — tier 2 local coupling via κ_ω and stellar mass, unchanged.
- **V.D257, V.D258**: acceleration scales and BTFR — tier 2, unchanged.

No registry entries are retracted. Five Ch. 62/63 text updates are recommended for v2.3.2 (listed in §8).

Appendix B: glossary of new terminology

Tier 1 (global background). The cosmic-mean densities ω_m, ω_b, ω_r, Ω_Λ carried by the D-sector of $H_\partial[\omega]$ as structural properties of the holonomy content. Set by V.T198 (M3h) and V.T200.

Tier 2 (local matter-phase coupling). The coupling weight of a local matter phase $\mathcal{P}$ (stars, cold HI, hot ICM, photon–baryon fluid) into the capacity-gradient source term in the rotation/lensing equation. Set by the SPARC-175 fit (V.T85 v2.3 form) and the Bullet-scoping limit. Weights are $\{1, \kappa_\omega, \approx 0\}$ for $\{D, \omega, B\}$-sector phases respectively.

Criterion 1. The two-tier sector-assignment rule: global densities live at tier 1 via D-sector holonomy; local-phase couplings live at tier 2 via the trinary decomposition. The two tiers are structurally decoupled.

Appendix C: data products

- `code/recombination_sector_analysis.py` — Python script producing the decoherence-rate table and the ℓ_1 sensitivity scenarios. Reproducible with Python 3 + stdlib only.
- `data/v4-5-summary.json` — machine-readable numerical summary (constants, Gamma/H ratios, ℓ_1 scenarios, Y_p note, chosen resolution).